



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Malaysia-Japan
International
Institute of Technology
(MJIIT)

USER MANUAL

Lab & Equipment Booking System

A comprehensive guide for reserving laboratory spaces and equipment at MJIIT, UTM

Application Development

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1.1 Login Interface

The Login Interface is the main entry point of the Lab Booking System. Users must enter valid credentials to access the system. The system verifies the email and password before granting access.

The diagram illustrates the Lab Booking System login interface. It features a central yellow box with a rounded border. At the top, the title "Lab Booking System" is displayed in a bold, dark red font, followed by the subtitle "Sign in with your university account" in a smaller, grey font. Below the subtitle, there are two input fields: "Email" and "Password". The "Email" field is a white rectangle with a red border, and the "Password" field is a white rectangle with a red border. Below the "Email" field, there is a red-bordered box containing the text "Use your @graduate.utm.my email address." with a red arrow pointing to it from the label "ii" on the left. Below the "Password" field, there is a red-bordered box containing the text "Forgot password?" with a red arrow pointing to it from the label "iv" on the right. Below the "Forgot password?" box, there is a large red button with the text "Login" in white. Below the "Login" button, there is a red-bordered box containing the text "Don't have an account? Register" with a red arrow pointing to it from the label "v" on the right. The label "i" is positioned to the right of the "Email" field with a red arrow pointing to it. The label "iii" is positioned to the right of the "Password" field with a red arrow pointing to it.

Functional Description: The core entry mechanism verifying user authenticity before granting session tokens to access the underlying lab database.

System Logic: The system executes a lookup queries against user registration tables. If the email is present and the salted/hashed password matches, a user session is initiated. If it fails, an error exception triggers.

Input Parameters & Examples:

- **Email String:** Must match an existing user entry.
 - *Example Input:* john.doe@graduate.utm.my
- **Password String:** A case-sensitive string matching the user registration block.
 - *Example Input:* P@ssw0rd2026!

b. Function Description

i. Email Input Field

Allows user to enter registered email for authentication.

ii. Password Input Field

Allows user to enter password for login verification.

iii. Forgot Password Link

Redirects user to password recovery page.

iv. Login Button

Validates user credentials and logs user into system.

v. Register Link

Redirects new user to registration page.

1.2 Sign Up / Create Account

The Create Account interface allows new users to register into the system using valid information. Users must select their role before creating an account.

The image shows a 'Lab Booking System' registration form with the following elements and annotations:

- Username** input field, annotated with **i**.
- Email** input field, annotated with **ii**. Below it is the text: 'Use your @graduate.utm.my email address.'
- Password** input field, annotated with **iii**.
- Role** dropdown menu with 'PIC' selected, annotated with **iv**. Below it is the text: 'PIC accounts are assigned by the system administrator after your account is created.'
- Register** button, annotated with **v**.
- Login** link (circled), annotated with **vi**. It is preceded by the text: 'Already have an account?'.

Functional Description: Creates a persistent user record assigning structural access scopes across specific institutional roles.

System Logic: Validates that the username and email are unique globally. The email domain must strict-match university requirements. Upon authorization, the record initializes with status flags corresponding to structural roles.

Input Parameters & Examples:

- **Username:** A unique string identification token.
 - *Example Input:* johndoe99

- **Email Address:** An institutional domain email format.
 - *Example Input:* smith@graduate.utm.my
- **Password Choice:** String enforcing strict corporate formatting complexities.
 - *Example Input:* Secure#Lab88
- **Role Allocation:** Enum variable defining specific permission flags within the system architecture.
 - *Example Input:* student (or choices of staff / pic)

b. Function Description

i. Username Field

Used to enter a unique username for system login.

ii. Email Field

Must be valid institutional email (@graduate.utm.my).

iii. Password Field

Used to create secure login password.

iv. Role Selection

Defines user access level (PIC / Staff / Student).

v. Register Button

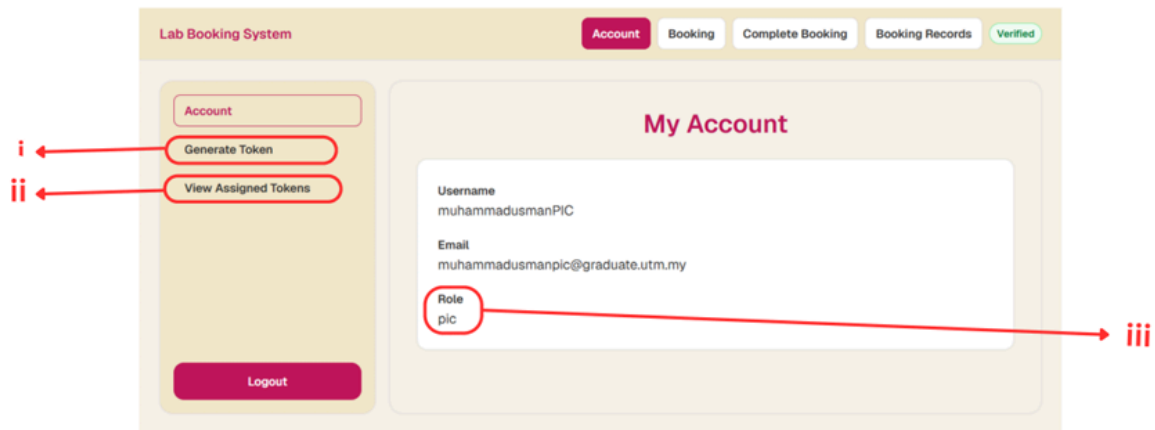
Submits registration form after validation.

vi. Login Link

Redirects to login page.

1.3 Account Interface (PIC Dashboard)

The My Account interface allows PIC users to view profile information and access token management functions.



System Explanation: Before generating a booking code, the Person-In-Charge (PIC) must verify that the student or staff member exists in the database. The system searches the user tables using the provided email string and returns the user's profile information if found.

Inputs and Examples:

- **User Email:** The exact university email address of the person requesting a token.
 - *Example:* `tifow@graduate.utm.my`

b. Function Description

i. Generate Token Button

Allows PIC to generate a 6-character booking token.

ii. View Assigned Tokens Button

Displays all tokens assigned to the user.

iii. User Profile Information

Displays username, email, and role information.

1.4 Token Generation Interface

The Token Generation Interface allows PIC users to generate and assign booking tokens to users for system access control.

The screenshot displays the 'Lab Booking System' interface. At the top, a navigation bar includes 'Account', 'Booking', 'Complete Booking', 'Booking Records', and 'Verified'. The left sidebar contains 'Account' with sub-links 'Generate Token' and 'View Assigned Tokens', and a 'Logout' button at the bottom. The main content area is titled 'Token Generation' with the subtitle 'Generate and assign booking tokens for students or staff.' It features three red-outlined boxes with arrows pointing to labels 'i', 'ii', and 'iii' on the right. Box 'i' contains the 'PIC workflow' (1. Search for the user by university email, 2. Generate a 6-character token, 3. Assign the token to that user, 4. The user enters it during lab or equipment booking). Box 'ii' contains a 'User email' input field and a 'Find user' button. Box 'iii' contains a 'Note' section stating the token is valid for 7 days and can be reused.

Lab Booking System

Account Booking Complete Booking Booking Records Verified

Account

Generate Token

View Assigned Tokens

Logout

Token Generation

Generate and assign booking tokens for students or staff.

PIC workflow

1. Search for the user by university email. 2. Generate a 6-character token. 3. Assign the token to that user. 4. The user enters it during lab or equipment booking.

User email

Find user

Note:

- The generated code is valid for 7 days.
- The user can reuse this code for multiple bookings within those 7 days.

i

ii

iii

b. Function Description

i. PIC Workflow Description Panel

Explains the steps of token generation process.

ii. User Email Input Field

Used to search user by email before generating token.

iii. Generate Code Button

Generates a 6-character token and assigns it to the user.

c. System Rules

- Token is valid for 7 days
- Token is required for booking access
- Only PIC can generate tokens
- Each token is unique per user

1.5 Token Assignment Result Interface

This interface shows the generated token details and confirms assignment to the user.

The diagram illustrates the 'Token Generation' interface. At the top, a header 'Token Generation' is followed by the subtitle 'Generate and assign booking tokens for students or staff.' Below this is a 'PIC workflow' section containing four steps: 1. Search for the user by university email, 2. Generate a 6-character token, 3. Assign the token to that user, and 4. The user enters it during lab or equipment booking. A red arrow labeled 'i' points to this workflow section. Below the workflow is a green status bar that says 'User found successfully.' Underneath is a form with a 'User email' field containing 'tifow@graduate.utm.my' and a red 'Find user' button. A red arrow labeled 'ii' points to the 'Find user' button. Below the form is a 'User details' section with a table showing 'Username' as 'tifow@graduate.utm.my', 'Role' as 'student', and 'Email' as 'tifow@graduate.utm.my'. Below the table is a red 'Generate code' button. A red arrow labeled 'iii' points to the 'Generate code' button.

Token Generation
Generate and assign booking tokens for students or staff.

PIC workflow
1. Search for the user by university email. 2. Generate a 6-character token. 3. Assign the token to that user. 4. The user enters it during lab or equipment booking.

User found successfully.

User email
tifow@graduate.utm.my

Find user

User details

Username	Role	Email
tifow@graduate.utm.my	student	tifow@graduate.utm.my

Generate code

b. Function Description

i. Generated Code Information

Displays token, expiry date, username, and role.

ii. Assigned Status Button

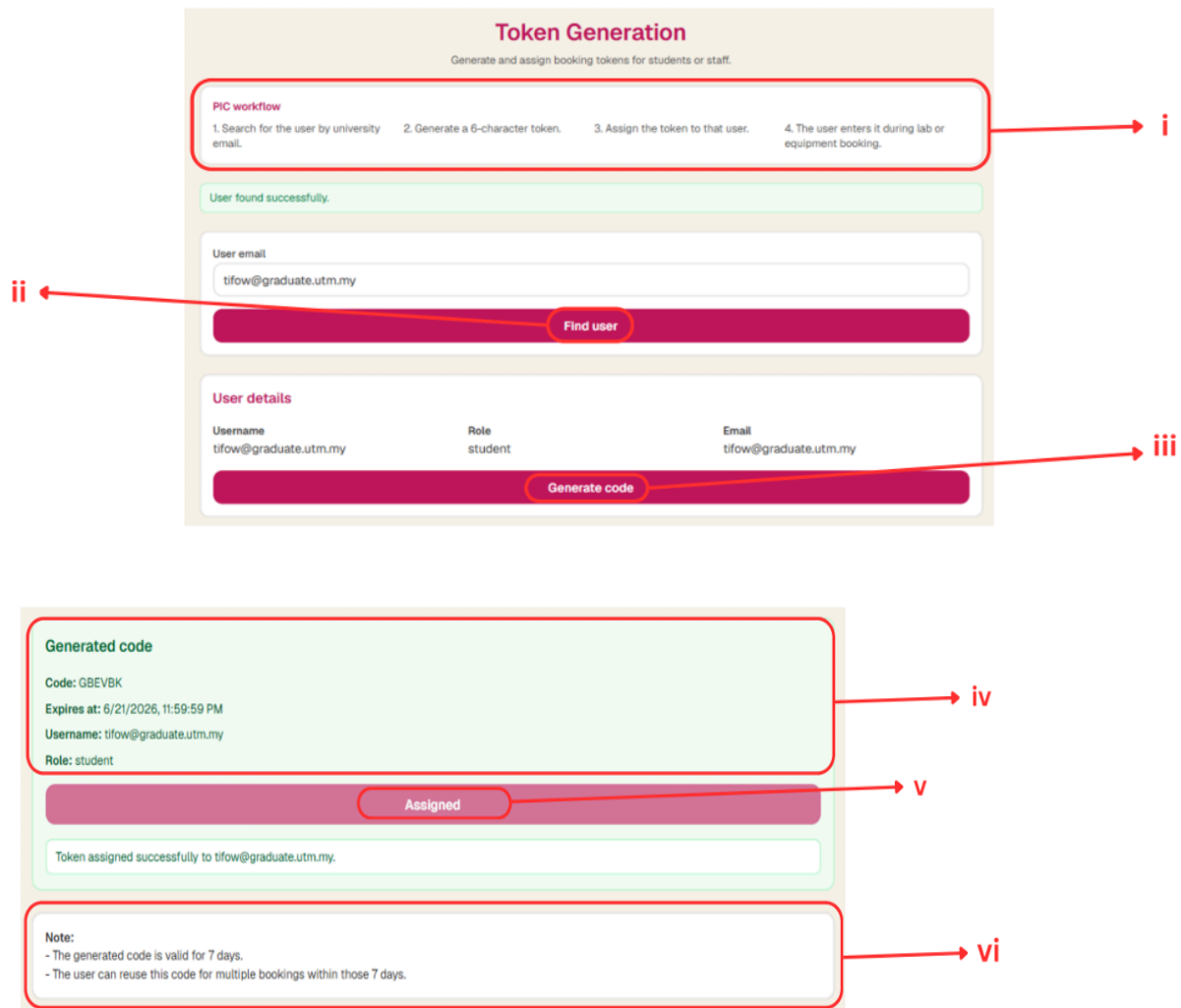
Indicates that token has been successfully assigned.

iii. Notes Section

Displays token validity rules and usage conditions.

1.6 Token Generation Interface

The Token Generation Interface allows PIC users to generate and assign booking tokens to users. This token is required for system access during lab or equipment booking.



The screenshot displays the 'Token Generation' interface, which is used to generate and assign booking tokens for students or staff. The interface is divided into several sections, with annotations (i through vi) pointing to specific elements:

- Token Generation** (Header): Generate and assign booking tokens for students or staff.
- PIC workflow** (Section i): A list of four steps: 1. Search for the user by university email. 2. Generate a 6-character token. 3. Assign the token to that user. 4. The user enters it during lab or equipment booking.
- User found successfully.** (Status message)
- User email** (Form field): Contains the email address 'tifow@graduate.utm.my'.
- Find user** (Button ii): A red button used to search for the user.
- User details** (Section iii): A table showing user information:

Username	Role	Email
tifow@graduate.utm.my	student	tifow@graduate.utm.my
- Generate code** (Button iii): A red button used to generate the token.
- Generated code** (Section iv): A green box containing the generated token details:
 - Code: GBEVBK
 - Expires at: 6/21/2026, 11:59:59 PM
 - Username: tifow@graduate.utm.my
 - Role: student
- Assigned** (Button v): A red button used to assign the token.
- Token assigned successfully to tifow@graduate.utm.my.** (Status message)
- Note:** (Section vi): A list of two notes:
 - The generated code is valid for 7 days.
 - The user can reuse this code for multiple bookings within those 7 days.

System Explanation: This creates a short, unique 6-character code that allows a verified user to book lab spaces and tools. The system saves this code in a tracking table and automatically sets an expiration time of exactly 7 days (168 hours) from the moment it was made.

System Rules:

- Only users with a **pic** account role can generate these tokens.
- The token can be used for multiple bookings until its 7-day period ends.

Inputs and Examples:

- **Selected User:** The specific user profile found during the search process.
 - *Example:* **USER-98231**

b. Function Description

i. PIC Workflow Information Panel

Displays the step-by-step process of token generation:

1. Search user by email
2. Generate a 6-character token
3. Assign token to user
4. User uses token for booking

ii. User Email Input Field + Find User Button

Allows PIC to search a user using their institutional email before generating a token.

iii. User Details Panel + Generate Code Button

Displays user information (username, role, email) and allows PIC to generate a booking token.

iv. Generated Code Information Panel

Displays the generated token details including:

- Token code
- Expiry date and time
- Username
- Role

v. Assigned Status Button + Confirmation Message

Indicates successful token assignment and displays confirmation message to PIC.

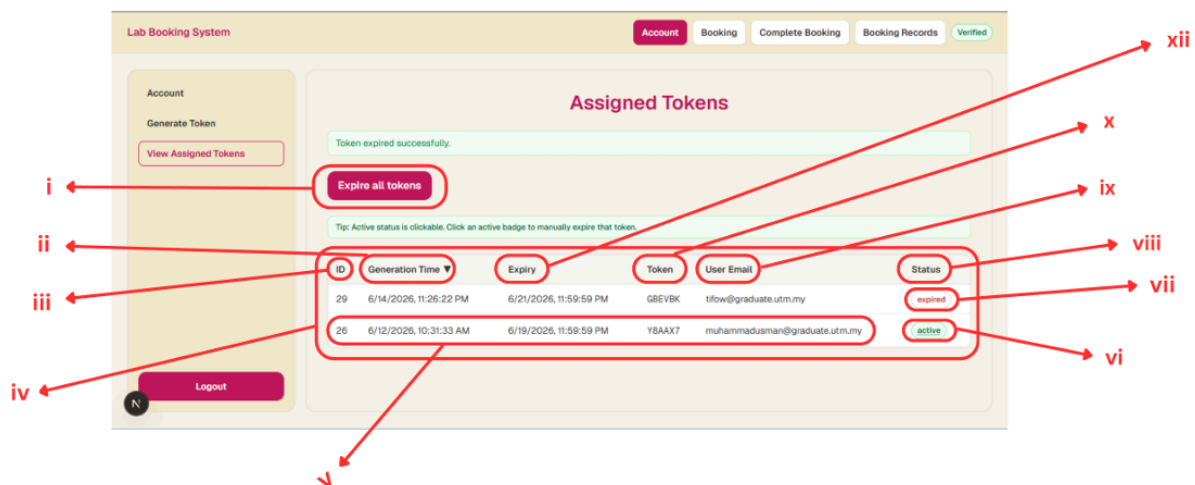
vi. Notes / System Rules Section

Provides system rules regarding token usage:

- Token is valid for 7 days
- Token can be reused for multiple bookings within validity period
- Token is required for booking access

1.7 Assigned Tokens Interface

The Assigned Tokens Interface allows PIC users to view, manage, and control all generated booking tokens. It provides information about token status, expiry, and user assignment.



Action i: Expire All tokens

- **System Explanation:** A master backend action that searches the database for all currently usable booking tokens and changes their status to invalid all at once. This instantly cuts off booking access for everyone holding a temporary token, without waiting for their individual 7-day timers to run out.

- **Input Parameters:** This action is a direct system command that does not require manual text input. It alters all active rows in the token storage table.
- **Example System Action:** The system finds all database entries marked as valid (such as token code **Y8AAX7**) and overwrites their status field to **expired**.

Data Field ii: Unique Record Identifier (Token ID)

- **System Explanation:** A unique database tracking number automatically assigned by the system to a token entry the moment it is created. This number ensures that every single token generated has its own permanent identity, preventing database mix-ups between different users or codes.
- **Input Parameter:** A whole number (Integer) generated by the database.
- **Example Input:** **26**

Data Field iii: Creation Timestamp (Generation Time)

- **System Explanation:** The exact date and time recorded by the server clock when a Person-In-Charge (PIC) successfully created the access token.
- **Input Parameter:** A standardized system date-and-time string.
- **Example Input:** **6/12/2026, 10:31:33 AM**

Data Structure iv: Token Information Row (Data Record)

- **System Explanation:** A complete database record object that holds all the individual pieces of information relating to a single token given to a user.
- **Input Parameter:** A grouped set of data entries containing the identification number, timestamps, token code, user identity, and current availability status.
- **Example Input Data:** A combined data entry consisting of: ID **26**, generated on **6/12/2026**, expiring on **6/19/2026**, code string **Y8AAX7**, assigned to user email **muhammadusman@graduate.utm.my**, with an **active** status value.

Data Field v: Cut-off Timestamp (Expiry Time)

- **System Explanation:** The specific future date and time when a token naturally becomes invalid. The system calculates this automatically by adding exactly 7 days (168 hours) to the original creation timestamp.
- **Input Parameter:** A calculated future date-and-time string.
- **Example Input:** **6/19/2026, 11:59:59 PM**

System State vi: (Active Status)

- **System Explanation:** A system state indicating that a token is currently open and valid. When a student or staff member attempts a booking using this token code, the system database approves the access verification check.
- **Input Parameter:** A text status text code or true/false status indicator.

- **Example Value:** `active`

System State vii: Invalid Token Flag (Expired Status)

- **System Explanation:** A system state indicating that a token can no longer be used because its 7-day lifespan has passed, or because a manager manually deactivated it. Any booking request containing this code will be rejected by the backend security system.
- **Input Parameter:** A text status text code or true/false status indicator.
- **Example Value:** `expired`

Data Field viii: Token State Tracker (Status Field)

- **System Explanation:** The specific data column in the system registry that monitors whether a token is currently usable or unusable.
- **Input Parameter:** A text selection matching the allowed system states.
- **Example Input:** `active`

Data Field ix: Authorized User Identity (User Email)

- **System Explanation:** The registered institutional email address belonging to the specific student or staff member assigned to the token. This ensures that a generated code can only be used by its rightful owner.
- **Input Parameter:** A text string following university email rule formats.
- **Example Input:** `muhammadusman@graduate.utm.my`

Data Field x: Alphanumeric Clearance Code (Token String)

- **System Explanation:** The actual 6-character randomized text code generated by the system backend that serves as a security passkey for equipment reservations.
- **Input Parameter:** A unique 6-character text code.
- **Example Input:** `Y8AAX7`

System Output xii: System Success Notification Message

- **System Explanation:** A backend confirmation message generated by the database engine to verify that a token status modification command successfully finished running without errors.
- **Input Parameter:** A simple feedback text string.
- **Example Output:** `Token expired successfully.`

b. Function Description

i. Sidebar Navigation Menu

Allows navigation between system modules such as Account, Generate Token, and View Assigned Tokens.

ii. Page Title

Displays the current page name “Assigned Tokens”.

iii. Expire All Tokens Button

Allows PIC to expire all active tokens in the system at once.

iv. Logout Button

Logs the user out of the system and ends the session.

v. Notification / Status Message Panel

Displays system messages such as successful token expiration or updates.

vi. Token Table Container

Displays all token records in a structured table format.

vii. Status Badge

Indicates token status:

- Green = Active
- Red = Expired

viii. User Email Column

Displays the email of the user assigned to the token.

ix. Token Column

Displays the generated 6-character token.

x. Expiry Column

Shows token expiration date and time.

xi. Generation Time Column

Shows when the token was created.

xii. ID Column

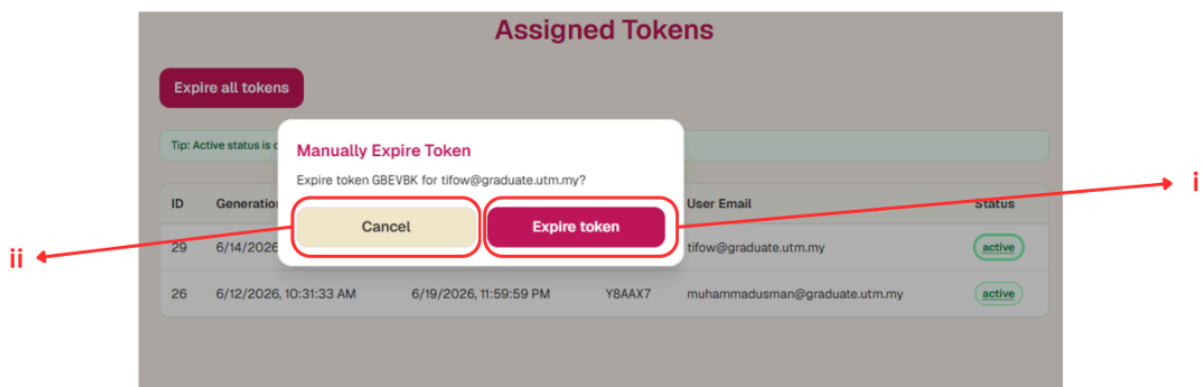
Displays the unique identifier for each token record.

c. System Rules

- Only PIC users can manage tokens
- Active tokens can be manually expired
- Expired tokens cannot be reused
- Each token is linked to a specific user
- Token status is updated in real time

1.8 Manual Token Expire Confirmation Dialog

This interface appears when the PIC user manually selects a token to expire. It is used to confirm the action before permanently changing the token status.



System Explanation: This allows a PIC to immediately cancel a token before its 7-day time limit naturally runs out. The system changes the token's status flag from active to

expired in the database. This can be done for one specific token or for all tokens in the system at the same time.

Inputs and Examples:

- **Token Code:** The exact 6-character text code that needs to be deactivated.
 - *Example:* **GBEVBK**

b. Function Description

i. Confirmation Message Panel

Displays the selected token information and asks the user to confirm whether they want to expire the token.

It ensures that the user does not accidentally expire a wrong token.

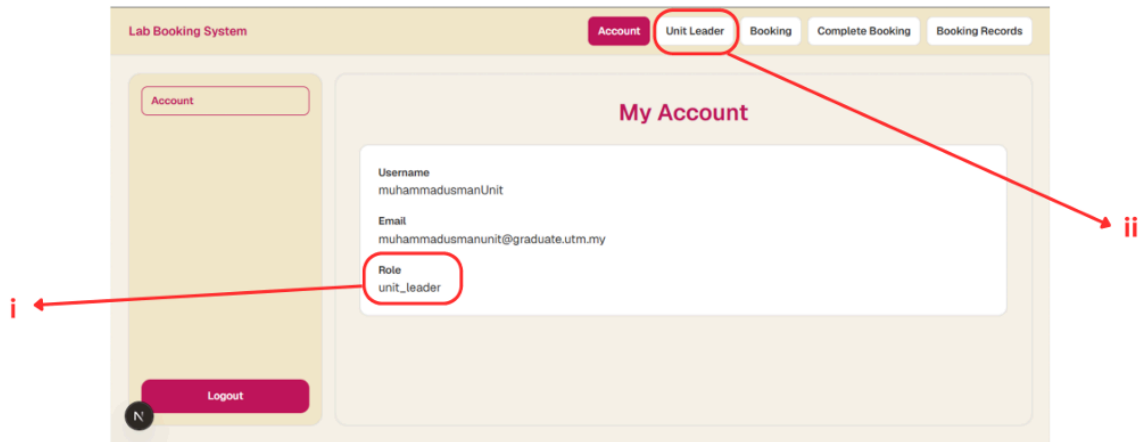
ii. Action Buttons

- **Cancel Button** → Closes the dialog without making any changes
- **Expire Token Button** → Confirms the action and changes token status to expired

c. System Behavior

- If “Cancel” is clicked → no changes are made
- If “Expire Token” is clicked → token status is updated to expired
- Action is irreversible once confirmed
- System updates token table automatically after action

2.0 Unit Leader Account Dashboard



b. The account dashboard shows the user's account information and role-based navigation.

i. **Role Information** – Displays the current role assigned to the logged-in user.
In this example, the user role is **unit_leader**.

ii. **Unit Leader Navigation Tab** – This tab appears because the logged-in user has the **Unit Leader** role.

It allows the user to access Unit Leader features and approval-related pages.

2.1 Unit Leader Recommendation Dashboard

The **Unit Leader Recommendation Dashboard** allows the Unit Leader to review booking requests before they are forwarded to PPMU for the final decision.

a. The Unit Leader dashboard page is shown below.

Lab Booking System

Account

Unit Leader

Booking

Complete Booking

Booking Records

Unit Leader Recommendation Dashboard

Review booking requests before they move to PPMU final decision.

How to review requests

Pending requests appear first. Use search and filters to narrow the list, open View Details to inspect the booking, then submit the unit leader recommendation.

7

Total Requests Received

0

Pending Review

4

Recommended

0

Rejected

Search by user name or request ID...

All Status

Clear Filters

i

ii

iii

iv

Lab Booking System

Account

Unit Leader

Booking

Complete Booking

Booking Records

Unit Leader Recommendation Dashboard

Review booking requests before they move to PPMU final decision.

How to review requests

Pending requests appear first. Use search and filters to narrow the list, open View Details to inspect the booking, then submit the unit leader recommendation.

7
Total Requests Received

0
Pending Review

4
Recommended

0
Rejected

Search by user name or request ID...

All Status

All Status

Pending

Partially Reviewed

Recommended

Rejected

v

REQUEST ID	USER NAME	STUDY LEVEL	LECTURER	REQUEST MADE	BOOKING DATE	ITEMS	PRICE
9	muhammadasman	UTM STUDENT/UNDERGRADUATE	fg gfgf@gmail.com gfgfg ID 1234	6/12/2026, 4:27 PM	2026-06-19	2	RM 1097.60
8	muhammadasman	UTM STUDENT/UNDERGRADUATE	Dr Aishah aishah@utm.my MJIT ID 23242	6/12/2026, 3:48 PM	2026-06-19	2	RM 548.80
7	zhawei	UTM STUDENT/UNDERGRADUATE	none none@graduate.utm MJ ID 1234	6/12/2026, 2:30 PM	2026-06-19	1	RM 156.80
6	muhammadasman	UTM STUDENT/UNDERGRADUATE	fg gfgf@gmail.com gfgfg ID 1234	6/12/2026, 12:25 PM	2026-06-19	1	RM 156.80
5	muhammadasman	Undergraduate	fg gfgf@gmail.com ID	6/12/2026, 10:47 AM	2026-06-19	2	RM 235.20

vi

vii

STUDY LEVEL	LECTURER	REQUEST MADE	BOOKING DATE	ITEMS	PRICE	UNIT LEADER RECOMMENDATION	ACTION
NT(UNDERGRADUATE)	Dr Zatul zatul@utm.my MJIT ID 1234	6/16/2026, 11:17 AM	2026-06-23	2	RM 431.20	Partially Reviewed	View Details
NT(UNDERGRADUATE)	Dr Aishah aishah@utm.my MJIT ID 12345	6/16/2026, 10:24 AM	2026-06-23	2	RM 2900.80	Partially Reviewed	View Details
NT(UNDERGRADUATE)	fg gfgf@gmail.com gfgfg ID 1234	6/12/2026, 4:27 PM	2026-06-19	2	RM 1097.60	Partially Reviewed	View Details
NT(UNDERGRADUATE)	Dr Aishah aishah@utm.my MJIT ID 23242	6/12/2026, 3:48 PM	2026-06-19	2	RM 548.80	Recommended	View Details
NT(UNDERGRADUATE)	none none@graduate.utm MJ ID 1234	6/12/2026, 2:30 PM	2026-06-19	1	RM 156.80	Recommended	View Details

viii

ix

b. The dashboard contains the following parts:

i. **Review Instruction Panel** – Displays a short guide explaining how the Unit Leader should review requests. Pending requests appear first. The Unit Leader can use the search and filters to narrow the list, open **View Details** to inspect the booking, and then submit the recommendation.

ii. **Request Summary Cards** – Shows the total number of requests and their current recommendation status, including **Total Requests Received**, **Pending Review**, **Recommended**, and **Rejected**.

iii. **Status Filter Dropdown** – Allows the Unit Leader to filter the booking request list based on request status.

iv. **Clear Filters Button** – Clears the selected search and filter values and returns the request list to its default view.

v. **Status Filter Options** – Displays the available status options, such as **All Status**, **Pending**, **Partially Reviewed**, **Recommended**, and **Rejected**.

vi. **Booking Request Table Header** – Displays the main booking request columns, such as **Request ID**, **User Name**, **Study Level**, **Lecturer**, **Request Made**, **Booking Date**, **Items**, and **Price**.

vii. **Booking Request Row** – Shows the details of a specific booking request submitted by a user.

viii. **Unit Leader Recommendation and Action Columns** – Shows the current Unit Leader recommendation status and the available action for each request.

ix. **View Details Button** – Allows the Unit Leader to open the selected booking request and review the full request details before making a recommendation.

2.2 Unit Leader Request Detail Page

The **Unit Leader Request Detail Page** allows the Unit Leader to view full booking request information before submitting a recommendation decision.

a. After clicking the **View Details** button from the Unit Leader Recommendation Dashboard, the system will display the selected request detail page.

The screenshot displays the 'Unit Leader Request Detail Page' within the 'Lab Booking System'. The page features a top navigation bar with links for 'Account', 'Unit Leader', 'Booking', 'Complete Booking', and 'Booking Records'. A 'Back' button is located at the top left of the main content area.

The main content area is divided into several sections, each highlighted with a red box and an annotation:

- Annotation i:** Points to a summary box at the top right showing '2 equipment items' and a 'Total' of 'RM 431.20'.
- Annotation ii:** Points to the 'USER DETAILS' section, which includes fields for USERNAME (muhammadusman), ROLE (staff), EMAIL (muhammadusman@graduate.utm.my), USER ID (A23MJ3008), FACULTY (MJIT), CONTACT (01126163816), and STUDY LEVEL (UTM STUDENT(UNDERGRADUATE)).
- Annotation iii:** Points to the 'LECTURER DETAILS' section, which includes fields for LECTURER NAME (Dr Zatul), LECTURER EMAIL (zatul@utm.my), LECTURER CONTACT (0112618478), FACULTY (MJIT), ID NUMBER (1234), and VOT NUMBER (1234).
- Annotation iv:** Points to the 'PIC DETAILS' section, which includes fields for PIC TOKEN (WHBS24), PIC NAME (muhammadusmanPIC), and PIC EMAIL (muhammadusmanpic@graduate.utm.my).
- Annotation v:** Points to the 'REQUEST DETAILS' section, which currently shows a dash '-'.



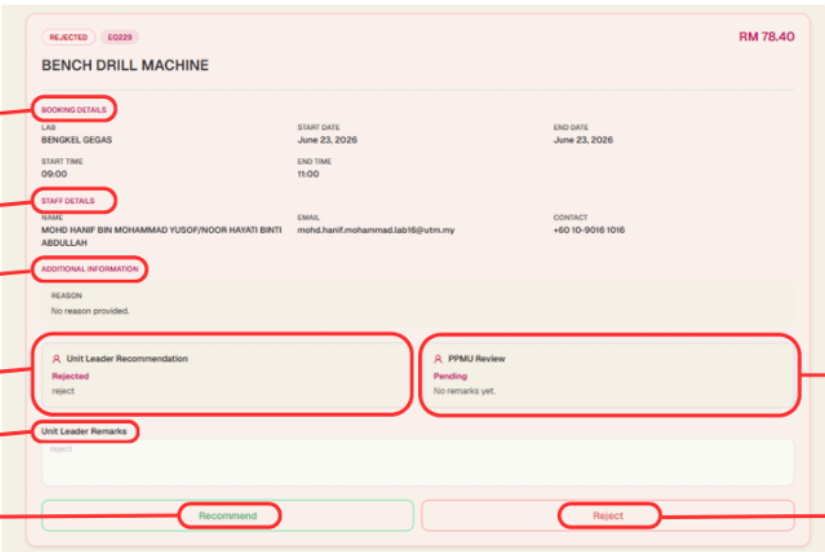
b. The request detail page contains the following parts:

- i. **Request Summary Header** – Displays the request number, current request status, number of equipment items, and the total estimated price of the booking request.
- ii. **User Details Section** – Displays the requester’s information, such as username, role, email, user ID, faculty, contact number, and study level.
- iii. **Lecturer Details Section** – Displays the lecturer information related to the booking request, including lecturer name, lecturer email, lecturer contact, faculty, ID number, and VOT number.
- iv. **PIC Details Section** – Displays the PIC information for the selected booking request, including PIC token, PIC name, and PIC email.
- v. **Request Details Section** – Displays the additional request information or remarks provided by the requester.
- vi. **Bulk Review Section** – Allows the Unit Leader to review available equipment items in the request together.
- vii. **Remarks for All Field** – Allows the Unit Leader to enter remarks that will be applied to every reviewed equipment item.
- viii. **Recommend All Button** – Allows the Unit Leader to recommend all available equipment items in the booking request.
- ix. **Reject All Button** – Allows the Unit Leader to reject all available equipment items in the booking request.

2.3 Equipment Item Review Section

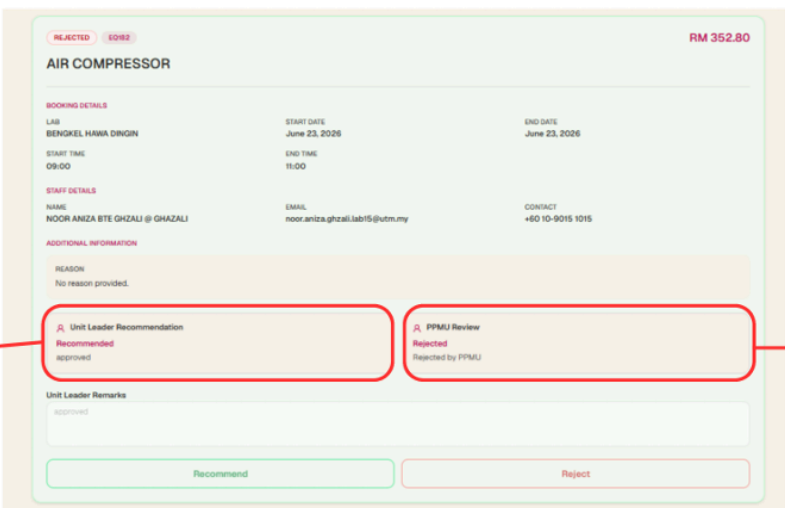
The **Equipment Item Review Section** allows the Unit Leader to review each equipment item inside a booking request before recommending or rejecting it.

a. The equipment item review card is shown below.



The image shows a review card for a 'BENCH DRILL MACHINE' with a price of RM 78.40. The card is divided into several sections, each labeled with a Roman numeral from i to viii. Section i points to the 'BOOKING DETAILS' header. Section ii points to the 'STAFF DETAILS' header. Section iii points to the 'ADDITIONAL INFORMATION' header. Section iv points to the 'Unit Leader Recommendation' box, which shows 'Rejected' and 'reject'. Section v points to the 'Unit Leader Remarks' box, which is empty. Section vi points to the 'Recommend' button. Section vii points to the 'Reject' button. Section viii points to the 'PPMU Review' box, which shows 'Pending' and 'No remarks yet.'.

REJECTED		E0279		RM 78.40	
BENCH DRILL MACHINE					
BOOKING DETAILS					
LAB	BENKOKEL GEGAS	START DATE	June 23, 2026	END DATE	June 23, 2026
START TIME	09:00	END TIME	11:00		
STAFF DETAILS					
NAME	MOHD HANIF BIN MOHAMMAD YUSOF/NOOR HAYATI BINTI ABDULLAH	EMAIL	mohd.hanif.mohammad.lab16@utm.my	CONTACT	+60 10-9016 1016
ADDITIONAL INFORMATION					
REASON	No reason provided.				
Unit Leader Recommendation			PPMU Review		
Rejected reject			Pending No remarks yet.		
Unit Leader Remarks					
reject					
Recommend			Reject		



The image shows a review card for an 'AIR COMPRESSOR' with a price of RM 352.80. The card is divided into several sections, each labeled with a Roman numeral from ix to x. Section ix points to the 'Unit Leader Recommendation' box, which shows 'Recommended' and 'approved'. Section x points to the 'PPMU Review' box, which shows 'Rejected' and 'Rejected by PPMU'.

REJECTED		E0182		RM 352.80	
AIR COMPRESSOR					
BOOKING DETAILS					
LAB	BENKOKEL HAWA DINGIN	START DATE	June 23, 2026	END DATE	June 23, 2026
START TIME	09:00	END TIME	11:00		
STAFF DETAILS					
NAME	NOOR ANIZA BTE GHZALI @ GHZALI	EMAIL	noor.aniza.ghzali.lab15@utm.my	CONTACT	+60 10-9015 1015
ADDITIONAL INFORMATION					
REASON	No reason provided.				
Unit Leader Recommendation			PPMU Review		
Recommended approved			Rejected Rejected by PPMU		
Unit Leader Remarks					
approved					
Recommend			Reject		

b. The equipment item review card contains the following parts:

i. **Booking Details Section** – Displays the booking information for the selected equipment, including lab name, start date, end date, start time, and end time.

ii. **Staff Details Section** – Displays the staff information related to the selected lab or equipment, including staff name, email, and contact number.

iii. **Additional Information Section** – Displays the reason or additional notes related to the equipment booking request.

iv. **Unit Leader Recommendation Status** – Shows the current Unit Leader recommendation result for the equipment item, such as **Rejected**.

v. **Unit Leader Remarks Field** – Displays or allows the Unit Leader to enter remarks related to the recommendation decision.

vi. **Recommend Button** – Allows the Unit Leader to recommend the selected equipment item.

vii. **Reject Button** – Allows the Unit Leader to reject the selected equipment item.

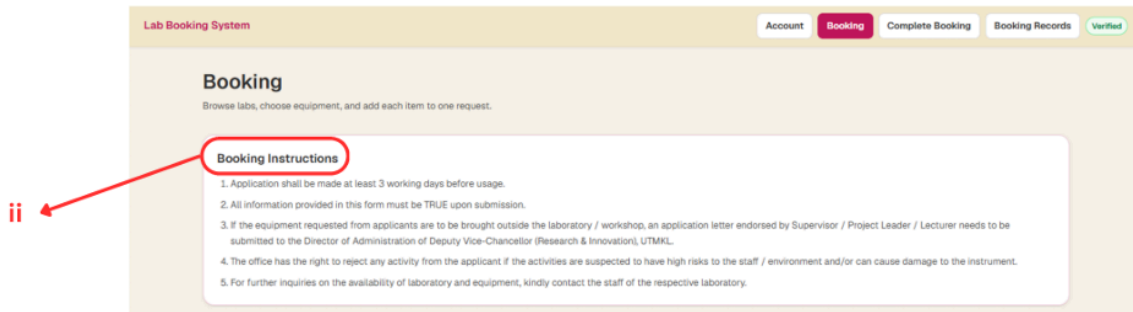
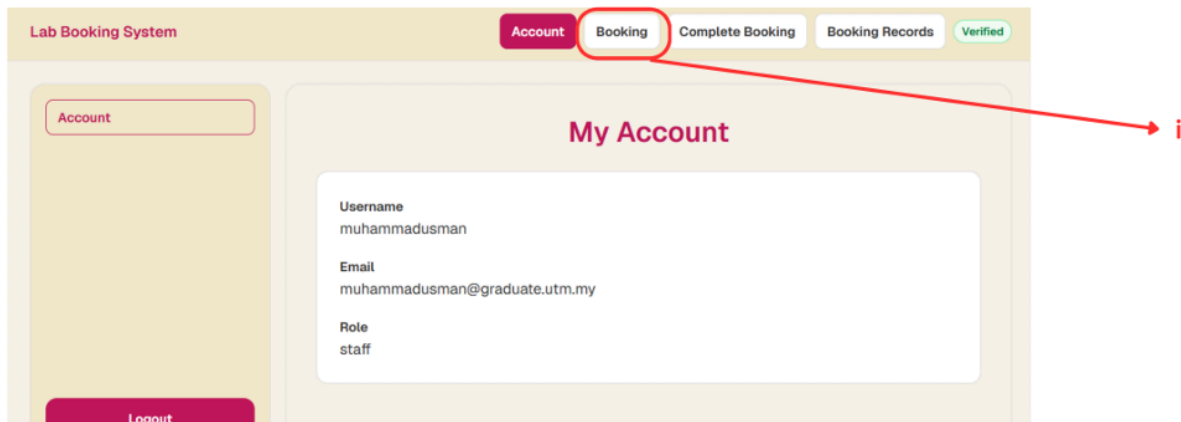
viii. **PPMU Review Status** – Shows the current PPMU review status for the selected equipment item, such as **Pending**.

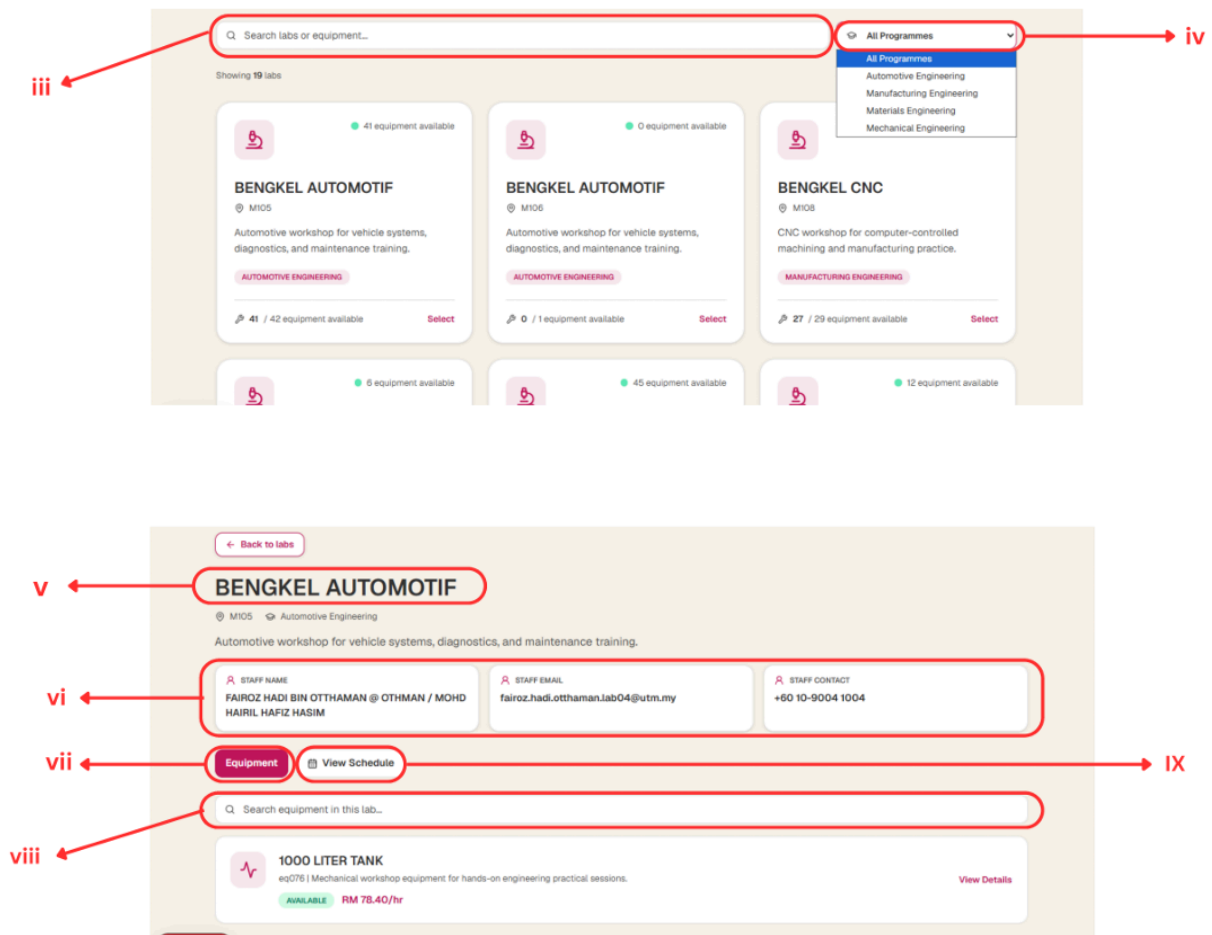
ix. **Recommended Status Example** – Shows an example where the Unit Leader has recommended the equipment item and added remarks.

x. **PPMU Rejection Status Example** – Shows an example where the equipment item has been rejected by PPMU, including the PPMU rejection remark.

1.6 Booking Interface

The **Booking Interface** allows verified users to search laboratories, filter laboratories by programme, view lab information, check equipment provided by the lab, and continue the booking process. Users can access this page after logging in and clicking the **Booking** button from the navigation bar.





a. The Booking Interface contains the following parts:

i. **Booking Button** – Allows the user to open the booking page from the account dashboard. When the user clicks this button, the system redirects the user to the booking page where laboratories and equipment can be searched and selected.

Example: A staff user who wants to book equipment from **Bengkel Automotif** must click **Booking** first before selecting the laboratory or equipment.

ii. **Booking Instructions Panel** – Displays important rules that users must read before making a booking.

The instructions explain that the application should be made before the usage date, information must be correct before submission, and requests may be rejected if the activity is risky or unsuitable.

Example: If a user wants to use equipment next week, the user should check the booking instructions first to make sure the request follows the system rules.

iii. **Search Lab / Equipment Input Field** – Allows the user to search for a laboratory or equipment by typing a keyword.

The system will filter the displayed lab cards based on the entered search keyword.

Example: The user can type **Bengkel Automotif** to search for the automotive workshop.

Example: The user can type **1000 Liter Tank**, **Bench Drill Machine**, or **Air Compressor** to search for equipment.

iv. **Programme Filter Dropdown** – Allows the user to filter laboratories based on programme category.

The available options include **All Programmes**, **Automotive Engineering**, **Manufacturing Engineering**, **Materials Engineering**, and **Mechanical Engineering**.

Example: If the user selects **Automotive Engineering**, the system displays only laboratories related to the Automotive Engineering programme.

Example: If the user selects **All Programmes**, the system displays all available laboratories.

v. **Lab Name / Lab Title** – Displays the selected laboratory name.

This helps the user confirm that they are viewing the correct laboratory before selecting equipment.

Example: The selected lab shown is **BENGKEL AUTOMOTIF**.

vi. **Staff Information Section** – Displays the staff details for the selected laboratory.

This section normally includes the staff name, staff email, and staff contact number. Users can refer to this information if they need to ask about lab or equipment availability.

Example: Staff Name: **FAIROZ HADI BIN OTTHAMAN @ OTHMAN / MOHD HAIRIL HAFIZ HASIM**

Example: Staff Email: **fairoz.hadi.otthaman.lab04@utm.my**

Example: Staff Contact: **+60 10-9004 1004**

vii. **Equipment Tab** – Shows the list of equipment available inside the selected laboratory.

When this tab is selected, the system displays equipment names, availability status, and price per hour.

Example: Under **BENGKEL AUTOMOTIF**, the user can see equipment such as **1000 LITER TANK**.

viii. **Search Equipment in This Lab Input Field** – Allows the user to search only within the selected laboratory's equipment list.

This is useful when the lab has many equipment items and the user wants to find one quickly.

Example: The user can type **tank** to find **1000 LITER TANK**.

Example: The user can type **compressor** to find **Air Compressor**.

Example: The user can type **drill** to find **Bench Drill Machine**.

ix. **View Schedule Tab** – Allows the user to view the laboratory schedule.

This helps the user check whether the lab has available time slots before choosing equipment and submitting a booking request.

Example: Before booking equipment from **BENGKEL AUTOMOTIF**, the user can click **View Schedule** to check whether the lab is available on the required date.

b. After searching and selecting the required lab or equipment, the user can continue to view equipment details and add the selected equipment to the booking request.

1.8 View Equipment Details and Add to Request

The **Equipment Details** page allows the user to view information about a selected equipment item, check its weekly schedule, and add the equipment to a booking request. This page layout is the same for different equipment items, such as **Air Impact Wrench** and **Angle Grinder**. Only the equipment name, equipment code, price, and selected booking details may change.

a. After selecting an equipment item from the laboratory equipment list, the system displays the equipment detail page.

b. The equipment detail page contains the following parts:

i. **Equipment Name** – Displays the name of the selected equipment.

This helps the user confirm that they are booking the correct equipment before choosing a date and time.

Example: The selected equipment may be **AIR IMPACT WRENCH** or **ANGLE GRINDER**.

ii. **Weekly Schedule Section** – Shows the weekly schedule for the selected equipment.

The user can use this section to check whether the equipment has existing bookings or timetable blocks before submitting a request.

Example: If the user wants to book the equipment on **16/06/2026**, the user can view the weekly schedule for that week first.

iii. **Schedule Status Legend** – Explains the meaning of the schedule labels shown on the timetable.

The labels help the user understand whether the equipment is already requested, booked, or blocked by a class timetable.

Example:

Pending Request means another user has submitted a request that is waiting for approval.

Booked means the equipment has already been approved for use.

Class Timetable Booked means the equipment or lab is blocked for class usage.

iv. **Date Field** – Allows the user to select the date used to display the weekly schedule.

After selecting a date, the schedule will show the related week.

Example: The user can select **06/16/2026** to view the schedule for the week of **15 June 2026 – 21 June 2026**.

v. **Add to Request Section** – Allows the user to enter the required booking information before adding the equipment to the request basket.

This section is used after the user confirms that the equipment is suitable and available.

vi. **Start Date Field** – Allows the user to choose the first date of equipment usage.

The start date should match the date when the user wants to begin using the equipment.

Example: 06/23/2026.

vii. **Start Time Field** – Allows the user to choose the time when the equipment usage begins.

The selected start time must be earlier than the end time.

Example: 09:00.

viii. **Purpose / Reason Field** – Allows the user to explain why the equipment is needed.

The reason should be clear so the approver can understand the purpose of the request.

Example: FYP experiment testing, lab practical session, equipment demonstration, or mechanical workshop activity.

x. **Add to Request Button** – Adds the selected equipment and booking details into the request basket.

After clicking this button, the item is added to the current booking request. The user may continue selecting more equipment or proceed to complete the booking form.

Example: After entering **Start Date: 06/23/2026, Start Time: 09:00, End Time: 11:00,** and **Purpose: FYP experiment testing,** the user clicks **Add to Request.**

xi. **End Time Field** – Allows the user to choose the time when the equipment usage ends.

The end time must be after the start time.

Example: 11:00.

xii. **End Date Field** – Allows the user to choose the last date of equipment usage.

For same-day usage, the end date can be the same as the start date.

Example: If the equipment is only needed for one session, the start date and end date can both be **06/23/2026.**

c. The same steps apply to different equipment detail pages. For example, the user can follow the same process when booking **AIR IMPACT WRENCH, ANGLE GRINDER, 1000 LITER TANK,** or any other equipment listed under the selected laboratory.

Picture 1.8: Equipment Details and Add to Request Page

Lab Booking System

Account

Booking

Complete Booking

Booking Records

Verified

← Back to BENGKEL AUTOMOTIF



RM 78.40/hr

AIR IMPACT WRENCH

eq052 | BENGKEL AUTOMOTIF

Mechanical workshop equipment for hands-on engineering practical sessions.

STAFF NAME

FAIROZ HADI BIN OTTHAMAN @ OTHMAN / MOHD
HAIRIL HAFIZ HASIM

STAFF EMAIL

fairoz.hadi.otthaman.lab04@utm.my

STAFF CONTACT

+60 10-9004 1004

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Lab Booking System

Account

Booking

Complete Booking

Booking Records

Verified

← Back to BENGKEL AUTOMOTIF



RM 78.40/hr

ANGLE GRINDER

eq053 | BENGKEL AUTOMOTIF

Mechanical workshop equipment for hands-on engineering practical sessions.

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STAFF CONTACT

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ii

iii

Weekly Schedule

Jun 15 - Jun 21, 2026

Date

06/16/2026

Previous Week

Next Week

PENDING REQUEST

BOOKED

CLASS TIMETABLE BOOKED

Mon Jun 15	Tue Jun 16	Wed Jun 17	Thu Jun 18	Fri Jun 19	Sat Jun 20	Sun Jun 21
No blocks	No blocks	No blocks	No blocks	No blocks	No blocks	No blocks

IV

V ← **Add to Request**

VI ← **Start Date** 06/23/2026 **End Date** 06/23/2026 Xii

VII ← **Start Time** 09:00 **End Time** 11:00 Xi

VIII ← **Purpose / Reason**
Test

X ← **Add to Request**